

Ques. HOW ARE ROOT-MEAN SQUARE (RMS) AND PEAK MUSIC POWER OUTPUT (PMPO) USED TO REPRESENT THE POWER OF AN AUDIO AMPLIFIER?

Anirvan Kule

Ans. A similar question was raised by a reader much earlier and the answer was published in EFY (August 1998). The same answer is reproduced below.

RMS. It is the minimum continuous sinewave power output measured in watts that can be delivered by the amplifier into a rated load, over the specified bandwidth with maximum specified total harmonic distortion.

PMPO. This is the capability of an amplifier either over a short duration during which there is no perceptible sag of the power supplies during the measurement, or when very high-quality power supplies are used with normal music input. It appears that some companies simply multiply RMS output by 8 or $(2\sqrt{2})^2$. RMS power is a more relied upon method for comparison.

Even though PMPO is not a well-defined standard of measurement, it is often seen on the product packaging or on the panel of a car audio amplifier (Fig. 1). Different manufacturers use different definitions for this rating. PMPO is more of a marketing tool to attract customers who are looking for a higher power rating with lower-cost audio amplifiers.

Q2. WHAT IS MACHINE LEARNING? WHICH PYTHON LIBRARIES ARE AVAILABLE FOR IT?

Pamarthi Kanakaraja

A2. Machine learning, or ML, is an application of artificial intelligence (AI) that provides systems the ability to

automatically learn and improve from experience without being explicitly programmed. It is based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention.

ML algorithms and models are used in a wide variety of applications such as email filtering, computer visions, self-driving cars and others. The types of ML algorithms depend on the task or problem intended to be solved, approached, and the type of input and output. Four

the software and so on.

Semi-supervised learning. This algorithm represents the middle ground between supervised and unsupervised algorithms. Use-cases may be found in legal and healthcare industries, among others, to manage Web content classification, image and speech analysis.

Reinforcement learning. This learning represents the actual ML AI. Applications of reinforced learning algorithms are found in self-driving cars and AlphaGo programs.

Python is one of the most popular ML programming languages. It is suitable for ML algorithms because it has an easy syntax. Python has many libraries for various applications including ML. For example, SciPy and Numpy are great for linear algebra and getting to know kernel methods of ML. Other common libraries are Scikit-learn, Theano, TensorFlow, Keras, PyTorch, Pandas and Matplotlib.

For more information on Python libraries, you may visit the

following websites:

<https://www.geeksforgeeks.org/best-python-libraries-for-machine-learning/>

<https://medium.freecodecamp.org/essential-libraries-for-machine-learning-in-python-82a9ada57aeb>

<https://medium.com/activewizards-machine-learning-company/top-15-python-libraries-for-data-science-in-in-2017-ab61b4f9b4a7>

popular types of ML algorithms are described below.

Supervised learning. This algorithm involves direct supervision of the operation. The most common application of this algorithm is price prediction and trend forecasting in sales, retail commerce and stock trading. The algorithm uses incoming data to assess the possibilities and calculate possible outcomes.

Unsupervised learning. This algorithm does not involve direct control of the programmer. It is used in digital marketing and ad technology to identify target audience based on certain credentials including behavioural data, personal data, customised settings in



Fig. 1: An example of PMPO rating on the panel of a car audio amplifier

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